INCH - POUND

MIL-L-46811B (MI)
10 September 1990
SUPERSEDING
MIL-L-46811A (MI)
27 October 1969

MILITARY SPECIFICATION

LUBRICATION OF GROUND HANDLING, LAUNCHING AND CONTROL EQUIPMENT FOR GUIDED MISSILES

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This process specification covers the requirements for lubrication of ground handling, launching, and control equipment for guided missiles in the course of manufacture and assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standard, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1440

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SPECIFICATIONS

FEDERAL

TT-S-1732	Sealing Compound; Pipe Joint and Thread, Lead Free General Purpose

MILITARY

MIL-L-2104	Lubricating Oil, Internal Combustion Engine, Tactical Service (Metric)
MIL-G-4343	Grease, Pneumatic System
MIL-H-5606	Hydraulic Fluid, Petroleum Base: aircraft, Missile, & Ordnance
MIL-G-10924	Grease, Automotive and Artillery
MIL-C-26373	Corrosion Preventive Compound, Solvent Cutback, Cold Application
MIL-G-23827	Grease, Aircraft and Instrument Gear and Actuator Screw, NATO Code Number G-354, Metric
MIL-L-46167	Lubricating Oil, Internal Combustion Engine, Arctic

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue. Philadelphia, PA 19111-5094.

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The lubricants shall successfully meet the material requirements o-f the applicable specification specified herein.

3.2 Equipment. Lubricant guns, cans, brushes, storage containes, and allied equipment shall be kept clean and free from contaminants.

3.3 Surface preparation. All dirt moisture, or other contamination on surfaces to be lubricated shall be removed prior to the application of the lubricant.

3.3.1 Cleaning. Prior to the application of lubricants or preservatives, assemblies and components shall be free of corrosion and residues foreign to the item. Critical functioning o close tolerance surfaces shall be cleaned by an appropriate process to insure removal of corrosion. If practical, internal parts of complex assemblies shall be cleaned prior to assembly, and precautions taken to avoid contamination thereafter.

3.4 Application. Where practical, components shall be operated while being Lubricated to assure full coverage of bearing surfaces. Lubricants

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shall be applied to threaded parts when necessary to prevent seizure or galling during assembly or installation of such parts.

3.4.1 Fluid line fittings. Lubricants, where required, shall be applied sparingly to the male threads. Lubricant shall not be applied on the nose of a fitting, the first 1.5 turns of thread from the nose, or the female threads. Lubricant shall not be allowed to enter the fluid system.

3.4.2 Acid oxidizer systems. The lubricant used on the acid oxidizer system and in elements of pneumatic systems associated therewith shall be one of the polymers of trifluorovinyl chloride as specified.

WARNING

Oil or grease shall not beapplied to fittings of an oxygen container, nor allowed to come in contact with an oxygen system.

3.4.3 Fuel systems. When lubricating any associated components of the fuel systems, a lubricant conforming to MIL-G-4343 or TT-S-1732 shall be used.

3.4.4 Compressed air system. Any lubricant used shall conform to MIL-G-4343.

3.4.5 Hydraulic fluid system. The lubricant used shall conform to MIL-H-5606 as applicable for the system.

3.4.6 Other threaded parts. Lubricants where required, shall be applied on the male threads only. On threads requiring movement or frequent adjustment, lubricant conforming to MIL-G-10924 shall be used. Where loading or occasional adjustment is required, lubricant conforming to TT-S-1732 shall be used. For sub-zero applications, lubricant conforming to MIL-L-46167 shall be used.

3.5 Grease fittings. Grease shall be forced into fittings until grease is extruded. The extruded surplus grease shall be wiped off withlint free wipers.

3.6 Friction bearings. Friction bearings (bushings) of steel or brass shall be lubricated with grease conforming to MIL-G-10924 or oil conforming to MIL-L-2104.

3.6.1 Antifriction bearing. Antifriction bearings (with rolling components) shall be lubricated using grease conforming to MIL-G-23827.

3.7 O-rings. Hydraulic system O-rings shall be lubricated by the dip, and drain method prior to installation: Specified lubricants for the various applications and contacts shall be used on the O-rings.

3.8 Specific parts and assemblies. The equipment and components assembled in the course of manufacture or rebuild shall be lubricated in accordance with tables I and II.

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3.9 Workmanship. Workmanship shall be such as to insure compliance with the requirements of this specification and shall be of good quality.

TABLE I. Lubrication list for launching equipment.

ITEM	Lube	Method	Application
Running gear units, wheel bearings-hand pack with grease.	GGA	Hand	On assembly
Hand brake levers, latches, catches, and shaft-turning points - oil lightly, remove excess	OE	Can	At install
Base leg assembly attach points - oil lightly, wipe off excess	OE	Can	At install
Jacks (mechanical), housing-case telescoping surfaces - raise jack as high as possible and apply light coat of grease to face of housing and case.	GAA	Hand	On assembly
(1) Remove cap and apply to grease fitting.	GAA	Gun	At install
(2) Base floating-ball joint - apply light coat of grease.	GAA	Hand	On assembly
(3) Bevel pinion-shaft - oil lightly.	OE	Can	On assembly
Hydraulic actuators			
(1) Piston erector attach point - apply to grease fittings.	GAA	Gun	At install
(2) Cylinder piston felt wipers, oil lightly - wipe off excess.	OHA	Can	On assembly
(3) Piston rod - apply preservative lightly.	CPH	Brush	On assembly
Erector rails and beam assemblies			
(1) Hinge joint - apply to grease fitting.	GAA	Gun	At install
(2) Connecting pins and locks - oil lightly, remove excess	OE	Can	On assembly
(3) Locating pin for launcher rail outer face and shaft slotted holes - oil lightly, remove excess.	OE	Can	At install
(4) Hydraulic connector hinge points and spring control release shaft - oil lightly.	OE	Can	At install
(5) End support and stop support hinge, hold down screw, lock latch release mechanism and hold open latch, turning points - oil lightly.	OE	Can	At install

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TABLE I. Lubrication list for launching equipment - Continued.

Item	Lube	Method	Application
(6) Rail assembly wheels - grease fittings in each wheel.	GAA	Gun	At install
(7) Rail assembly lateral guides - oil lightly.	OE	Can	At install
(8) Rail erecting spring lock pin and handle assembly sliding surfaces - oil lightly.	OE	Can	At install
(9) Locks manually operated, screw and trunnion, level slots, torque tube bearings, torque tube bell crank, plunger slot and plunger surfaces - oil lightly	OE	Can	At install
(10) Stabilizer jacks and legs, friction clamp, foot pad hinges or pins, leg attaching points - oil lightly.	OE	Can	At install
(11) Attaching pins, base floating ball joint - apply light coat of grease by hand.	GAA	Hand	On assembly
(12) Spring lock pin and handle assembly sliding surfaces - oil lightly.	OE	Can	At install
(13) Locks manually operated, screws and trunnions, lever slots, torque tube bell crank, plunger slot and surfaces - oil lightly.	OE	Can	At install
Foot brakes, turning points and threads - oil lightly.	OE	Can	At install
Fixed casters and bearing swivels-oil lightly or apply grease to fittings.	OE GAA	Can Gun	At install At install
Pintle bars, turn tables or fifth wheels - apply grease in an applicable manner.	GAA	Gun	On assembly
Cable drum bearings - pack or grease fittings as required.	GAA	Gun	On assembly
(1) Brake adjust nut - oil lightly	OE	Can	On assembly
(2) Cable - light film of preservative	CPH	Brush	On assembly
(3) Exposed surfaces of finished shafts, pins, etc. - apply a light film of corrosion preventive compound.	CPH	Brush	On assembly

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TABLE II. Lubricants, hydraulic fluids and corrosion preventatives.

Military Symbol	Military Specification	Description	Availability
OE	MIL-L-2104	Lubricating oil, Internal Combustion Engine, Tactical Sercice (Metric)	Commercially available to Specification
GPS	MIL-L-4343	Grease, Pneumatic System	Commercially available to Specification
OHA	MIL-H-5606	Hydraulic Fluid, Petroleum Base: Aircraft, Missile and Ordnance	Commercially available to Specification
OE	MIL-L-46167	Lubricating Oil, Internal Combustion Engine, Arctic	Commercially available to Specification
GAA	MIL-G-10924	Grease, Automotive and Artillery	Commercially available to Specification
CPH	MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application	Commercially available to Specification
GIA	MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric	Commercially available to Specification

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise speified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 4. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material. either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Materials. Evidence of the compliance of each lubricant to the respective specification as listed in 2.1.1 shall be available for inspection by the contracting officer.

4.3 Inspection. Each unit or component lubricated in accordance with this specification shall be inspected to verify fulfillment of specified requirements.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the requirements for lubrication of guided missile launching and handling equipment, as applicable, at the time of manufacture or assembly. Requirements for equipment lubrication in an operational status are covered in applicable lubricaiton orders (LOs).

6.2 Aquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2.1).

6.3 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.4 Subject term (keyword) listing.

Grease
Greasing
Lubricant
Oiling

6.5 Changes from previous issue. Marginal notations are not used in this revision to denote changes-with respect to the previous issue due to the extensiveness of the changes.

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-46811B (MI)	10 September 1990

3. DOCUMENT TITLE
LUBRICATION OF GROUND HANDLING, LAUNCHING AND CONTROL EQUIPMENT FOR GUIDED MISSILES

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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